

SIMATS ENGINEERING

ASSESSMENT TOOL 1

COURSE NAME: CSA16 COURSE CODE: Data Warehousing and Data Mining

COURSE OUTCOME COVERED

CO1: Design data warehouse architectures with appropriate dimensional models for effective decision support systems. BL1,BL2
Assessment Tool Weightage: 40%

QUESTIONS: 4

Total Marks:20

Annexure A
MCQ TEST

Criteria	Conceptual Understanding(3)	Accuracy of Answer(2.5)	Application of Knowledge(2)	Completeness of Response(1.5)	Clarity & Presentation(1)	Total(10)
Weightage	30%	25%	20%	15%	10%	100%
Q1						
Q2						
Q3						
Q4						

Code of Conduct:

I G SAI DEEPAK(192311432) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: Deepak G

Inside the Data Cube

A Journey Through OLAP Operations

How analysts roll up, drill down, slice, dice, and pivot their way to insight

G SAI DEEPAK

Reg No: 192311432

Course: CSA1603 – Data Warehouse and Data Mining



What Is OLAP?

One dataset, viewed from every angle

Online Analytical Processing lets business users explore data stored as a **multidimensional cube** — not a flat spreadsheet.

- **Dimensions** — Time, Product, Region — the ways you slice the business
- **Measures** — Sales, Profit, Quantity — the numbers you're analyzing
- **Hierarchies** — Day → Month → Year, or City → State → Country



THE DATA CUBE

X – Time

Y – Product

Z – Region

Every query touches multiple dimensions at once — giving answers a flat table never could.

Roll-Up & Drill-Down

Climbing and descending the concept hierarchy



ROLL-UP

Aggregates detailed data by climbing up a concept hierarchy, or by reducing the number of dimensions — turning granular rows into a summarized view.

City

State

Country

summarized →



DRILL-DOWN

The reverse of roll-up — navigates from summarized data toward finer detail, revealing the lower levels of a hierarchy on demand.

Year

Quarter

Month

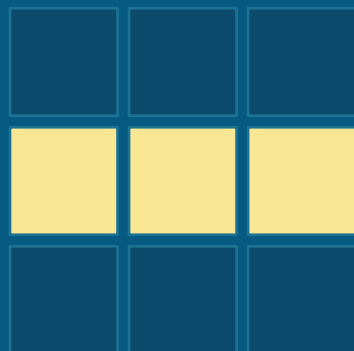
→ more detail

Slice & Dice

Carving a smaller, sharper view out of the cube

SLICE

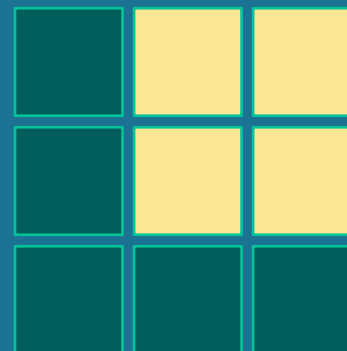
Fixes one dimension to a single value, cutting a single flat “slice” out of the cube — e.g. Sales where Year = 2025.



Year = 2025 → one full slice selected

DICE

Fixes two or more dimensions to specific values, producing a smaller sub-cube — e.g. Sales where Region = South AND Product = Electronics.



Region + Product fixed → sub-cube selected

Pivot (Rotate)

Same data, a completely new perspective



Pivot rotates the data axes of the cube to display an alternative orientation — for instance, swapping what appears in the rows versus the columns of a report. The underlying data never changes, only the vantage point.

Analysts use pivot to compare the same figures across different report layouts without re-querying the warehouse.

BEFORE — rows = Region

Product → columns	
North	
South	
East	

AFTER — rows = Product

Region → columns	
Laptops	
Phones	
Tablets	

Why it matters

A single pivot can turn a Region-first report into a Product-first report in seconds — no rebuilding the query, no waiting on IT.

From Raw Data to Real Decisions

Why OLAP operations matter in the real world



Seconds

to re-view data — not days of re-querying



Multi-Angle

insight from one cube, endless perspectives



Faster

business decisions, backed by live detail

Roll-up, drill-down, slice, dice, and pivot are the five moves that turn a static warehouse into a living conversation with the data — every rotation of the cube reveals a question the business didn't know it could ask.